Name:- Slock Ahuja **Enrollment no.**92400133041

Use appropriate smoothing and sharpening techniques to enhance the clarity of the images. **CO3, CO4**









Name:- Slock Ahuja **Enrollment no.**92400133041



Apply multiple filters, observe the changes produced by each method, and document your observations in detail.
Discuss how smoothing and sharpening individually affect image quality, and summarize your conclusions based on the results.

1st image:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


filename = '1st.png'


sharpen_kernel = np.array([[-1, -1, -1],

                           [-1,  9, -1],

                           [-1, -1, -1]])


img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)


if img is None:

   print(f"Error: Could not load {filename}. Please check the file path and
extension.")

else:

   gaussian_blur = cv2.GaussianBlur(img, (5, 5), 0)



   median_blur = cv2.medianBlur(img, 5)
```

```
sharpened = cv2.filter2D(img, -1, sharpen_kernel)



plt.figure(figsize=(10, 8))



plt.subplot(2, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original - 1st Image")
plt.axis('off')



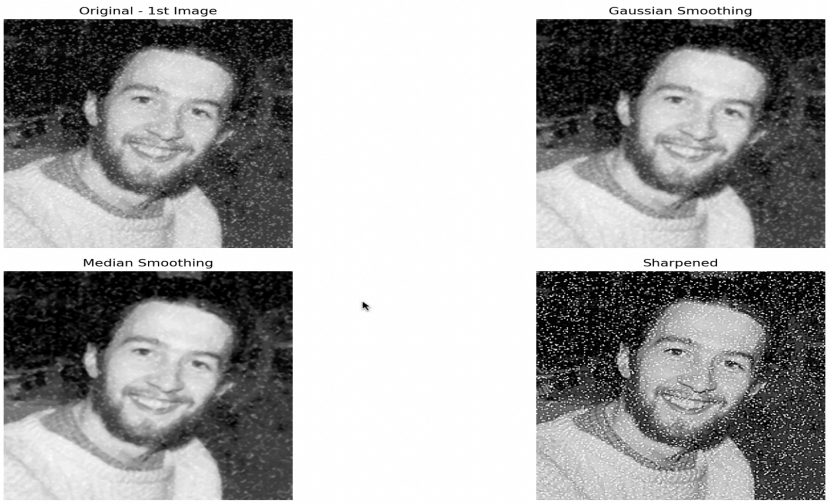
plt.subplot(2, 2, 2)
plt.imshow(gaussian_blur, cmap='gray')
plt.title("Gaussian Smoothing")
plt.axis('off')

plt.subplot(2, 2, 3)
plt.imshow(median_blur, cmap='gray')
plt.title("Median Smoothing")
plt.axis('off')

plt.subplot(2, 2, 4)
plt.imshow(sharpened, cmap='gray')
plt.title("Sharpened")
plt.axis('off')

plt.tight_layout()
```

```
plt.show()
```



Observation:-

Gaussian Smoothing (5x5 Kernel):
Reduces the sharp intensity of noise specks, but transforms them into a soft, hazy gray fog. Fine structural features (eyes, teeth, hairline) lose sharpness and appear visibly blurred/unfocused.

Median Smoothing (5x5 Kernel):
Most effective method for this specific image. It cleanly strips away the salt-and-pepper noise while keeping facial contours and major edges sharp, resulting in a significantly cleaner portrait without sacrificing detail.

Sharpening Filter (3x3 Laplacian-style Kernel):

Highlighting edges causes the filter to treat the noise specks as high-frequency detail. It drastically exaggerates the noise, making the image look harsh, heavily pixelated, and visually worse than the original.

2nd image:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


filename = '2nd.png'


sharpen_kernel = np.array([[-1, -1, -1],

                           [-1,  9, -1],

                           [-1, -1, -1]])
```

```
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)


if img is None:
   print(f"Error: Could not load {filename}. Please check the file path and
extension.")
else:


   gaussian_blur = cv2.GaussianBlur(img, (5, 5), 0)


   median_blur = cv2.medianBlur(img, 5)


   sharpened = cv2.filter2D(img, -1, sharpen_kernel)


   plt.figure(figsize=(10, 8))


   plt.subplot(2, 2, 1)
   plt.imshow(img, cmap='gray')
   plt.title("Original - 2nd Image")
   plt.axis('off')


   plt.subplot(2, 2, 2)
   plt.imshow(gaussian_blur, cmap='gray')
   plt.title("Gaussian Smoothing")
   plt.axis('off')


   plt.subplot(2, 2, 3)
   plt.imshow(median_blur, cmap='gray')
   plt.title("Median Smoothing")
   plt.axis('off')
```

```
plt.subplot(2, 2, 4)

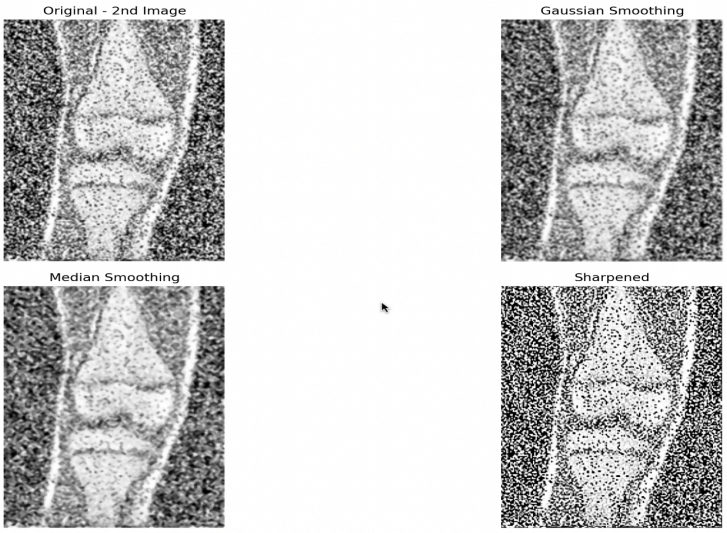
plt.imshow(sharpened, cmap='gray')

plt.title("Sharpened")

plt.axis('off')


plt.tight_layout()

plt.show()
```



Observations:-

Gaussian Smoothing (5x5 Kernel):

Highly effective for this image. It suppresses the high-frequency speckle noise efficiently, giving the medical scan a continuous, softer gray gradient. This makes tissue boundaries and bone contours easier to evaluate without harsh graininess, though very fine internal bone details get slightly softened.

Median Smoothing (5x5 Kernel):

Removes isolated noise spikes and preserves the main outer skeletal silhouette effectively. However, it can produce a blocky/patchy texture inside region gradients compared to Gaussian smoothing.

Sharpening Filter (3x3 Kernel):

Direct sharpening without prior smoothing amplifies the speckle noise aggressively. The overall background becomes very grainy, and false high-contrast artifacts appear inside the bone tissue, reducing diagnostic visual quality.

3rd image:-

```
import cv2

import numpy as np
```

Name:- Slock Ahuja **Enrollment no.**92400133041

```
import matplotlib.pyplot as plt


filename = '3rd.png'


sharpen_kernel = np.array([[-1, -1, -1],
                           [-1,  9, -1],
                           [-1, -1, -1]])


img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)


if img is None:
   print(f"Error: Could not load {filename}. Please check the file path and
extension.")
else:


   gaussian_blur = cv2.GaussianBlur(img, (5, 5), 0)


   median_blur = cv2.medianBlur(img, 5)


   sharpened = cv2.filter2D(img, -1, sharpen_kernel)


   plt.figure(figsize=(10, 8))


   plt.subplot(2, 2, 1)
   plt.imshow(img, cmap='gray')
   plt.title("Original - 3rd Image")
   plt.axis('off')

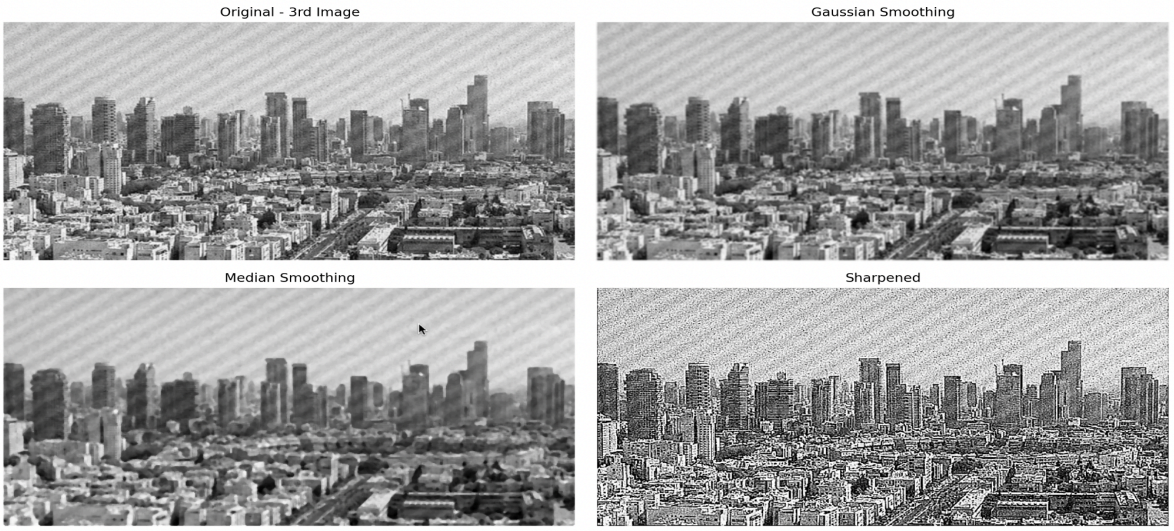

   plt.subplot(2, 2, 2)
   plt.imshow(gaussian_blur, cmap='gray')
```

```
plt.title("Gaussian Smoothing")
plt.axis('off')

plt.subplot(2, 2, 3)
plt.imshow(median_blur, cmap='gray')
plt.title("Median Smoothing")
plt.axis('off')

plt.subplot(2, 2, 4)
plt.imshow(sharpened, cmap='gray')
plt.title("Sharpened")
plt.axis('off')

plt.tight_layout()
plt.show()
```



Observations:-

Gaussian Smoothing (5x5 Kernel):

Softens distinct building edges and blurs fine architectural details like windows and structural frames, degrading overall image clarity.

Median Smoothing (5x5 Kernel):

Erodes fine structural features and sharp corners, giving building outlines a blocky, unnatural appearance and erasing distant architectural details.

Sharpening Filter (3x3 Kernel):

Delivers the best visual outcome. Since the image lacks heavy random noise, sharpening crisply highlights building outlines, windows, and sky/skyline boundaries without creating visible noise artifacts.

4th image:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


filename = '4th.png'


sharpen_kernel = np.array([[-1, -1, -1],

                           [-1,  9, -1],

                           [-1, -1, -1]])


img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)


if img is None:

   print(f"Error: Could not load {filename}. Please check the file path and
extension.")

else:

   gaussian_blur = cv2.GaussianBlur(img, (5, 5), 0)



   median_blur = cv2.medianBlur(img, 5)
```

```
sharpened = cv2.filter2D(img, -1, sharpen_kernel)


plt.figure(figsize=(10, 8))


plt.subplot(2, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original - 4th Image")
plt.axis('off')

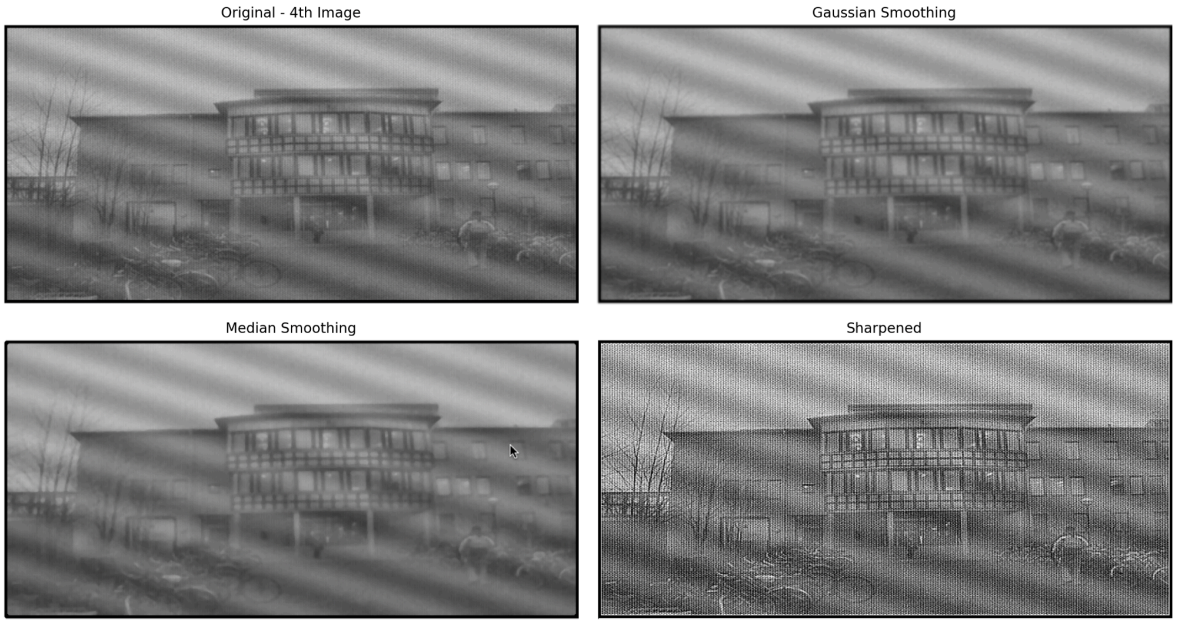

plt.subplot(2, 2, 2)
plt.imshow(gaussian_blur, cmap='gray')
plt.title("Gaussian Smoothing")
plt.axis('off')


plt.subplot(2, 2, 3)
plt.imshow(median_blur, cmap='gray')
plt.title("Median Smoothing")
plt.axis('off')


plt.subplot(2, 2, 4)
plt.imshow(sharpened, cmap='gray')
plt.title("Sharpened")
plt.axis('off')


plt.tight_layout()
plt.show()
```

Name:- Slock Ahuja **Enrollment no.**92400133041



Observations:-

Gaussian Smoothing (5x5 Kernel):

Blurs and diffuses the harsh boundaries between the horizontal stripes. It successfully suppresses high-frequency line noise and creates smoother transitions across adjacent bands, though at the expense of crisp line definition.

Median Smoothing (5x5 Kernel):

Heavily distorts parallel thin lines that fall below the kernel size. The regular stripe geometry becomes patchy or stepped, creating blocky artifacts along the horizontal stripe boundaries.

Sharpening Filter (3x3 Kernel):

Exaggerates the contrast between light and dark stripes. While it makes the horizontal edges look extremely crisp and defined, it also risks introducing harsh ringing artifacts along the stripe borders.

Name:- Slock Ahuja **Enrollment no.**92400133041

5th image:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt



filename = '5th.png'


sharpen_kernel = np.array([[-1, -1, -1],

                           [-1,  9, -1],

                           [-1, -1, -1]])


img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)


if img is None:

   print(f"Error: Could not load {filename}. Please check the file path and
extension.")

else:


   gaussian_blur = cv2.GaussianBlur(img, (5, 5), 0)


   median_blur = cv2.medianBlur(img, 5)


   sharpened = cv2.filter2D(img, -1, sharpen_kernel)


   plt.figure(figsize=(10, 8))


   plt.subplot(2, 2, 1)

   plt.imshow(img, cmap='gray')

   plt.title("Original - 5th Image")
```

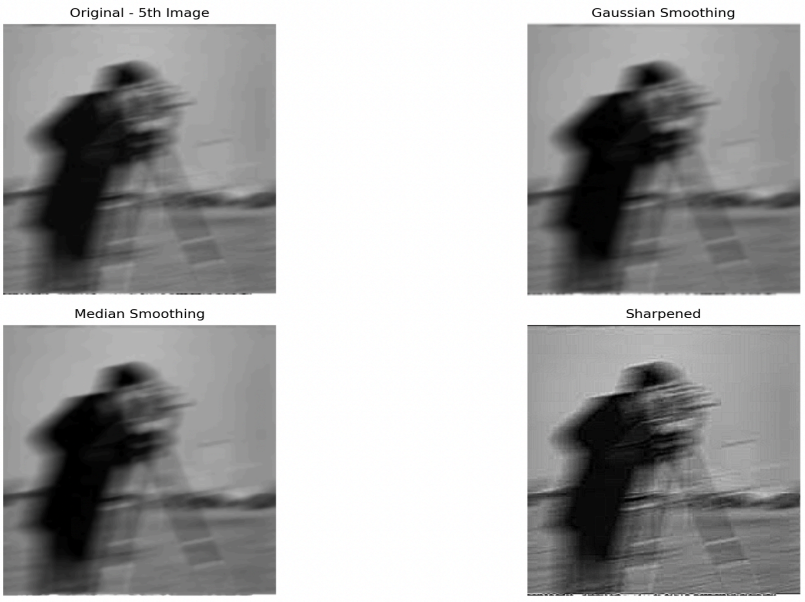
```
plt.axis('off')


plt.subplot(2, 2, 2)
plt.imshow(gaussian_blur, cmap='gray')
plt.title("Gaussian Smoothing")
plt.axis('off')


plt.subplot(2, 2, 3)
plt.imshow(median_blur, cmap='gray')
plt.title("Median Smoothing")
plt.axis('off')


plt.subplot(2, 2, 4)
plt.imshow(sharpened, cmap='gray')
plt.title("Sharpened")
plt.axis('off')


plt.tight_layout()
plt.show()
```



Name:- Slock Ahuja **Enrollment no.**92400133041

Observations:-

Gaussian Smoothing (5x5 Kernel):

Exacerbates the existing blur. Fine elements, such as the tripod legs and figure outline, blend into a soft, indistinct gradient blob, significantly reducing visual clarity.

Median Smoothing (5x5 Kernel):

Erases thin structures like the tripod legs and smooths out subtle shading gradients, turning the subject into a blocky, featureless silhouette.

Sharpening Filter (3x3 Kernel):

Delivers the most noticeable improvement in edge definition. By boosting high-contrast boundaries, it crispens the silhouette of the photographer and tripod against the sky, making the main subject stand out far better than in the original image.

GITHUB LINK:

https://github.com/SlockAhuja/DSIP.git